

Jiwon Heo



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Education

Korea University

Bachelor

2019 March - 2024 August

Seoul, Republic of Korea

- **Physics:** 1st Major, GPA 4.03/4.50
- **Mathematics:** Double Major, GPA 4.21/4.50
- **Economics:** Minor, GPA 4.08/4.50
- Total GPA 4.07/4.50

Experience

Signal Processing & Advanced Intelligence (SPAI) LAB (Prof. Jiwon Yoon)

July 2022 – December 2023

Intern

Seoul, Republic of Korea

- Organized "Accelerating secure permutation: Application to matrix algebra", wherein the contents focus on accelerating multiple operations within a homomorphic encryption scheme by leveraging its algebraic properties, serving as the primary author.
- Spearheaded requisite experiments in " ϕ -zer: Kernel optimizer for efficient computation over homomorphic encryption scheme" as co-author.
- Applied for a domestic patent titled "캐노니컬 임베딩을 사용하는 원본다항식몫체 기반의 암호문에서 특정 원소를 빠르게 추출하는 장치 및 방법".
- Participated the project with Statistics Korea, developing several statistical algorithms over homomorphic encryption scheme utilizing "OpenFHE".

Paper list

Accelerating secure permutation: Application to matrix algebra

Main author

Preprint

- Proposed a new, rapid method for multiplying ciphertext by a specialized form of plaintext, leveraging the algebraic properties of ciphertext rather than traditional homomorphic multiplication
- Exhibited theoretical and experimental improvements and availability, by applying to matrix multiplication.

ϕ -zer: Kernel optimizer for efficient computation over homomorphic encryption scheme

Co-author

IEEE transactions on information forensics and security (Expected)

- Accelerated various data analysis algorithms over homomorphic encryption scheme, including PCA and LDA, by utilizing kernel method.
- Contributed to the development of necessary algorithms, design, and execution of experiments and discussions, focusing on LDA and basic statistical algorithms.

Patent list

캐노니컬 임베딩을 사용하는 원본다항식몫체 기반의 암호문에서 특정 원소를 빠르게 추출하는 장치 및 방법

Domestic patent (Application no. 10-2023-0124483) (Proceeding)

- Proposed a new efficient operation about extracting specific elements from ciphertext based on cyclotomic polynomial quotient field, utilizing its algebraic properties.
- Led proactively patent application process, including drafting transmittal letter and meeting with patent attorney.

Core projects

Numerical analysis for ODE | Python

PHYS233

- Conducted numerical analysis of ordinary differential equations corresponding to classical dynamics with friction.

Hypothesis testing based on ML and its applications to physics experiments | Python, Tensorflow

PHYS466

- Designed and implemented hypothesis testing based on machine learning and apply to real experimental data.

Implementation of hypothesis testing | Python

PHYS303

- Embodied t-test and applied it to real data in order to determine whether experimental data can explain its corresponding theory.

Implementation of Bayesian statistics | Python

STAT404

- Implemented basic concepts of Bayesian statistics and applied them to real data.